

1 Solution — GIAM 1.7

Problem 4

1.1 Problem

Draw a five-pointed star and label all 10 points. There are 40 triples of these labels that satisfy the betweenness relation. List them.

1.2 The Star and Its Labels

A five-pointed star (pentagram) has 10 points: the 5 outer **tips** and the 5 inner **crossing points** (the vertices of the small pentagon in the middle). Label the tips A, B, C, D, E around the outside and the inner points F, G, H, I, J :

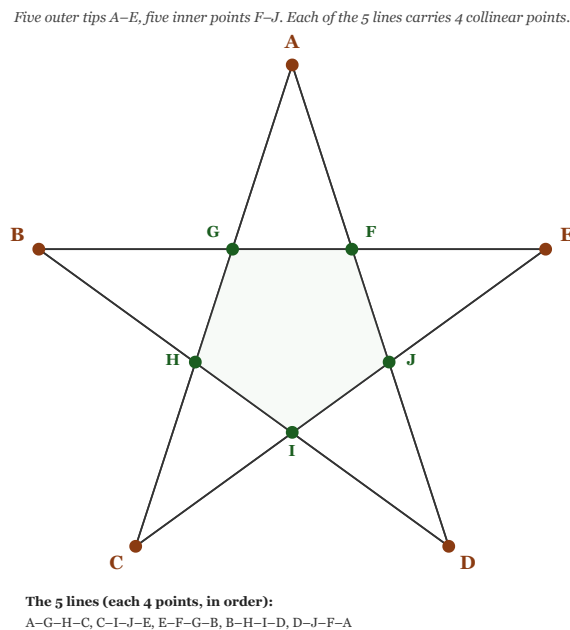


Figure 1.1: A five-pointed star drawn with five straight lines. Outer tips: A at top, B upper-left, C lower-left, D lower-right, E upper-right. Inner crossing points forming the central pentagon: F upper-right, G upper-left, H left, I bottom, J right. The five straight lines, each carrying four collinear points in order, are A–G–H–C, C–I–J–E, E–F–G–B, B–H–I–D, and D–J–F–A.

The crucial structural fact: the star is made of **five straight lines**, and each line passes through **four** of the ten points — two outer tips at its ends and two inner points between them. In order along each line:

$$A-G-H-C, \quad C-I-J-E, \quad E-F-G-B, \quad B-H-I-D, \quad D-J-F-A.$$

1.3 Counting: Why Exactly 40

Betweenness triples only occur among **collinear** points, so we work one line at a time. On a line with four points in order $P_1 P_2 P_3 P_4$, a triple (X, Y, Z) satisfies betweenness when Y lies strictly between X and Z . The middle point must be P_2 or P_3 (the ends P_1, P_4 are never between anything):

- P_2 between: $\{P_1, P_2, P_3\}$ and $\{P_1, P_2, P_4\}$;
- P_3 between: $\{P_1, P_3, P_4\}$ and $\{P_2, P_3, P_4\}$.

That is 4 unordered configurations per line. Following the section's convention — where the betweenness example listed *both* (A, B, C) and (C, B, A) — each configuration counts in **both directions**, giving 8 ordered triples per line. With 5 lines:

$$5 \text{ lines} \times 8 \text{ triples} = \boxed{40 \text{ triples}}.$$

1.4 The 40 Triples

Grouped by the line they live on (order written $X?Y?Z$, meaning Y is between X and Z):

Line $A-G-H-C$:

$$(A, G, H), (H, G, A), (A, G, C), (C, G, A), (A, H, C), (C, H, A), (G, H, C), (C, H, G)$$

Line $C-I-J-E$:

$$(C, I, J), (J, I, C), (C, I, E), (E, I, C), (C, J, E), (E, J, C), (I, J, E), (E, J, I)$$

Line $E-F-G-B$:

$$(E, F, G), (G, F, E), (E, F, B), (B, F, E), (E, G, B), (B, G, E), (F, G, B), (B, G, F)$$

Line $B-H-I-D$:

$(B, H, I), (I, H, B), (B, H, D), (D, H, B), (B, I, D), (D, I, B), (H, I, D), (D, I, H)$

Line D – J – F – A :

$(D, J, F), (F, J, D), (D, J, A), (A, J, D), (D, F, A), (A, F, D), (J, F, A), (A, F, J)$

1.5 Remark — Why Each Line Contributes Exactly 8

The count "8 per line" is worth pinning down, because it is where the 40 comes from. For n collinear points there are $\binom{n}{3}$ ways to choose three of them, and *every* such choice has exactly one middle point — so it yields exactly one unordered betweenness relation, hence two ordered triples. With $n = 4$ points per line, $\binom{4}{3} = 4$ unordered relations and $2 \times 4 = 8$ ordered triples. The 5 lines share points but never share a *triple* (two points already determine a unique line), so there is no double-counting: $5 \times 8 = 40$ exactly. Had we used only 3 collinear points per line we would get $\binom{3}{3} \cdot 2 = 2$ each; the pentagram's 4-per-line structure is what makes the total come out to the stated 40.

1.6 Remark — Betweenness Is Symmetric, So Triples Come in Pairs

Every triple above appears next to its reverse — (A, G, H) with (H, G, A) , and so on — because " Y is between X and Z " says exactly the same thing as " Y is between Z and X ." The *middle* label is what carries the geometric content; the two ends may be swapped freely. This is why the 40 is an even number: the triples pair off into 20 genuinely distinct betweenness facts, each recorded twice for the two reading directions. (Contrast the divisibility and $\leq$ relations of Problems 1 and 3, which are *not* symmetric — there (a, b) and (b, a) say different things.)

1.7 Remark — Betweenness Is a Ternary Relation

Unlike divisibility, $\leq$, or $=$ — all **binary** relations, sets of *pairs* — betweenness is a **ternary** relation: a set of ordered *triples*. This is exactly the point the section makes by introducing it. It cannot be reduced to a relation between two objects, because "between"

is inherently about a middle and two ends. Higher-arity relations are drawn not as a grid of dots (as the binary relations of Problems 1 and 3 were) but must be listed as tuples, which is why this problem asks for an explicit list rather than a single relation symbol.

1.8 Remark — The Inner Points Are Never Endpoints of the Whole Line

Notice that in every line’s ordering, the two **outer tips** are the extreme points and the two **inner points** sit in the middle — e.g. on $A-G-H-C$, the tips A, C are the ends and the inner points G, H are interior. Consequently each inner point serves as the “between” (middle) label more often than a tip does. This reflects the geometry of a pentagram: the inner pentagon is where all the crossings happen, so its vertices are precisely the points that lie *between* pairs of others. Every one of the **20** distinct betweenness facts has an inner point (F, G, H, I , or J) as its middle — no tip is ever strictly between two other marked points.